All about Docker Commands

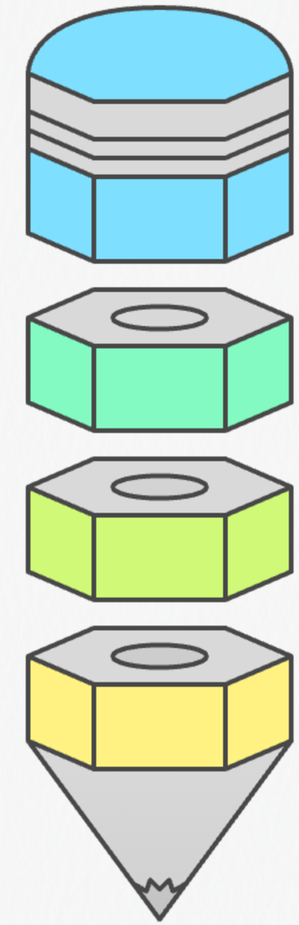
SILICON VALLEY ENGINEER

Image Management

Commands for handling Docker images

Container Interaction

Commands for interacting with containers



Container Management

Commands for managing Docker containers

Debugging Commands

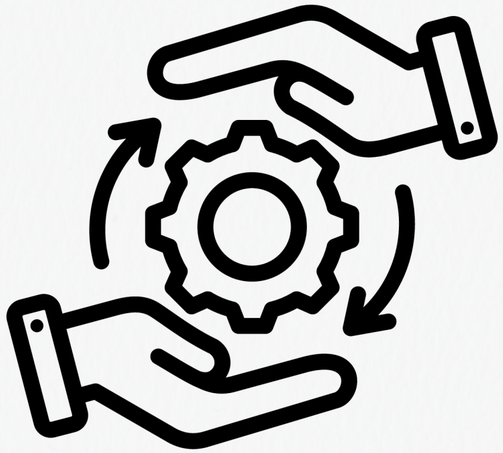
Commands for troubleshooting and debugging

General Docker Commands


```
# Displays the installed Docker version.
docker --version

# Shows system-wide information about Docker,
# including number of containers, images, and storage details.
docker info

# Lists all available Docker commands and options.
docker help
```

Image Management


```
# Lists all images on the local machine.
docker images

# Downloads an image from Docker Hub or another registry.
docker pull [image-name]

# Builds a Docker image using a Dockerfile in the current directory
# and tags it with a name.
docker build -t [image-name] .

# Tags an image with a different name or version.
docker tag [source-image] [target-image:tag]

# Pushes an image to a registry like Docker Hub.
docker push [image-name]

# Removes one or more images by ID or name.
docker rmi [image-id]
```



Container Management


```
# Lists all running containers.
docker ps

# Lists all containers, including stopped ones.
docker ps -a

# Creates and starts a container from an image
docker run [options] [image-name]
    # Common options
    -d # Run the container in detached mode.
    -p [host-port]:[container-port] # Maps a port from the host to the container.
    --name [container-name] # Assigns a name to the container.
    -v [host-path]:[container-path] # Mounts a volume.

# Removes a container.
docker rm [container-id/name]

# Forces the removal of a running container.
docker rm -f [container-id/name]
```

Container Management


```
# Stops a running container
docker stop [container-id/name]

# Starts a stopped container
docker start [container-id/name]

# Restarts a container
docker restart [container-id/name]
```

Container Interaction


```
# Runs a command inside a running container.
docker exec [container-id/name] [command]
# Example: docker exec -it [container-id] /bin/bash opens a bash shell in the container.

# Shows logs from a container.
docker logs [options] [container-id/name]
    --follow: Streams logs in real-time.

# Attaches your terminal to a running container's standard input/output.
docker attach [container-id/name]

# Forcing Immediate Termination. Sends SIGKILL
docker kill [options] [container-id/name]
    --signal, -s: Specifies the signal to send to the container. By default, it uses SIGKILL.
# Example signals: SIGTERM, SIGHUP, SIGINT.
```

Debugging Commands


```
# Shows detailed information about a container, image, or network.
docker inspect [object-id/name]

# Displays real-time resource usage statistics of running containers.
docker stats [container-id/name]

# Shows processes running inside a container.
docker top [container-id/name]

# Streams real-time events from the Docker daemon.
docker events
```